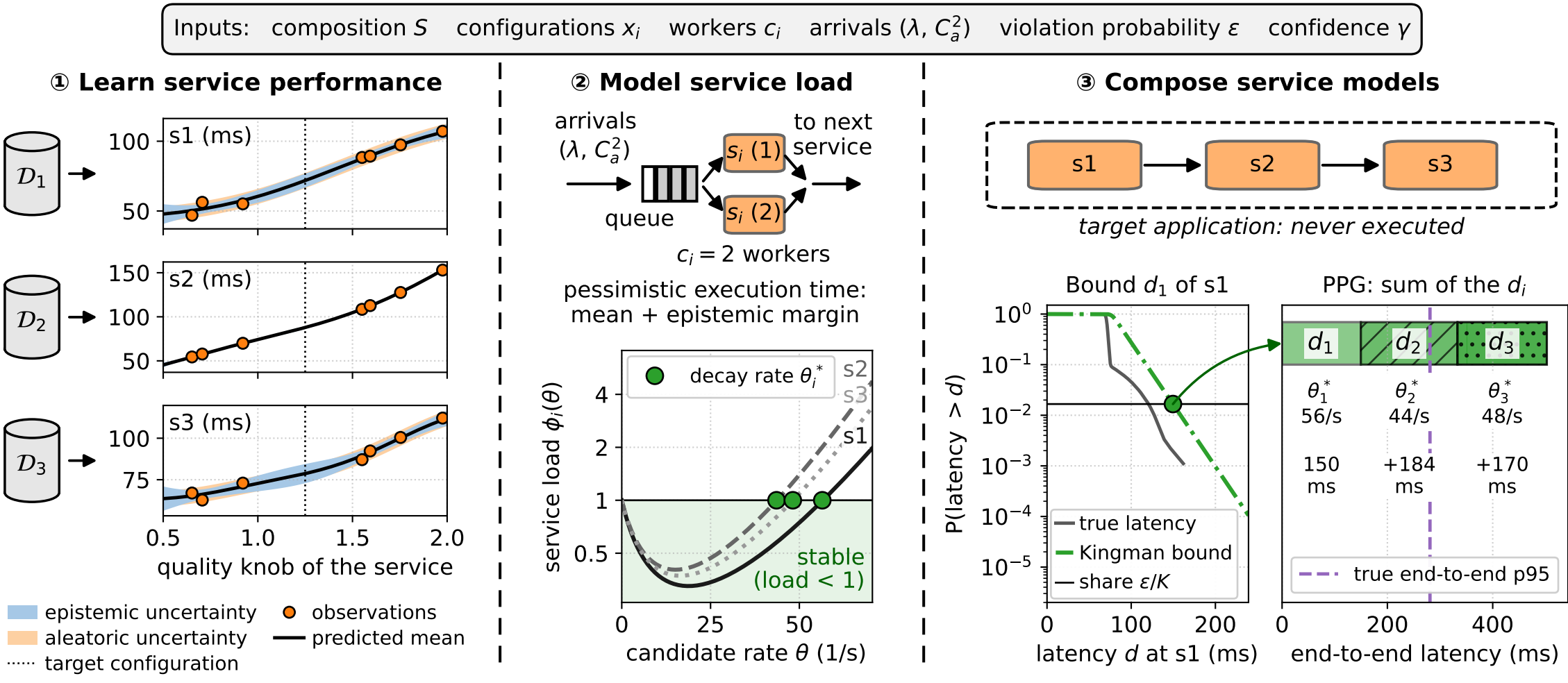

Inputs: composition S configurations x_i workers c_i arrivals (λ, C_a^2) violation probability ε confidence γ
① Learn service performance
$\mathcal{D}_1$
$\mathcal{D}_2$
$\mathcal{D}_3$
s1 (ms)
s2 (ms)
s3 (ms)
100
50
150
100
50
100
75
0.5
1.0
1.5
2.0
quality knob of the service
epistemic uncertainty
aleatoric uncertainty
target configuration
observations
predicted mean
② Model service load
arrivals
(λ, C_a^2)
queue
s_i (1)
s_i (2)
to next service
$c_i = 2$ workers
pessimistic execution time:
mean + epistemic margin
decay rate θ_i^*
s2
s3
s1
service load $\phi_i(\theta)$
4
2
1
0.5
stable
(load < 1)
0
25
50
candidate rate θ (1/s)
③ Compose service models
s1
s2
s3
target application: never executed
Bound d_1 of s1
PPG: sum of the d_i
P(latency > d)
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
true latency
Kingman bound
share ε/K
0
100
200
latency d at s1 (ms)
d_1
d_2
d_3
θ_1^*
56/s
θ_2^*
44/s
θ_3^*
48/s
150
ms
+184
ms
+170
ms
true end-to-end p95
0
200
400
end-to-end latency (ms)